Mesoporous Alumina-Silica via Hard Templating

ABSTRACT: Ordered mesoporous material was obtained by nanocasting using KIT-6 as the hard template.

2. EXPERIMENTAL

0.80 g of Pluronic F108 was dissolved in 40 mL of water with 2.5 mL of 1.0 M HNO3. Then 3.0 g of TMOS was added. The mixture was aged at 45 C for 12 h, treated hydrothermally at 130 C for 48 h, and calcined at 600 C for 4 h.

3. RESULTS

The calcined sample has a BET surface area of 430 m2/g and a BJH pore size distribution centred at a pore diameter of 11.5 nm. SEM shows plate-like particles with a hollow interior.